

# Method Metrics

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Effective Method Line Count

### Table 1a - Effective method line count distribution

This table shows the distribution of the effective method line count per artifact. For each artifact the number of methods with effective line count = 1,2,3,... is shown to get an overview of how line counts are distributed over methods.

Only the 15 artifacts with the highest method count and their effective method line count distribution (limited by 40) is shown here. The whole table can be found in the CSV report `Effective_Method_Line_Count_Distribution`.

Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName       | axon-messaging-5.0.3.jar | axon-common-5.0.3.jar | axon-server-connector-5.0.3.jar | axon-eventsourcing-5.0.3.jar | axon-test-5.0.3.jar | axon-modelling-5.0.3.jar | axon-spring-boot-autoconfigure-5.0.3.jar | axon-conversion-5.0.3.jar | axon-update-5.0.3.jar | axon-metrics-micrometer-5.0.3.jar |
|--------------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------------------|
| effectiveLineCount |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                                   |
| 1                  | 1933                     | 369                   | 266                             | 301                          | 212                 | 162                      | 186                                      | 116                       | 88                    | 46                                |
| 2                  | 524                      | 145                   | 143                             | 75                           | 73                  | 54                       | 74                                       | 22                        | 10                    | 18                                |
| 3                  | 424                      | 97                    | 77                              | 58                           | 36                  | 28                       | 16                                       | 16                        | 6                     | 20                                |
| 4                  | 191                      | 38                    | 44                              | 53                           | 40                  | 35                       | 6                                        | 8                         | 10                    | 8                                 |
| 5                  | 159                      | 36                    | 33                              | 31                           | 6                   | 22                       | 8                                        | 10                        | 8                     | 2                                 |
| 6                  | 93                       | 36                    | 15                              | 16                           | 15                  | 17                       | 8                                        | 7                         | 8                     | 4                                 |
| 7                  | 76                       | 15                    | 9                               | 14                           | 9                   | 9                        | 5                                        | 4                         | 2                     | 3                                 |
| 8                  | 68                       | 18                    | 15                              | 6                            | 8                   | 11                       | 2                                        | 6                         | 2                     | 4                                 |
| 9                  | 39                       | 11                    | 11                              | 10                           | 8                   | 12                       | 1                                        | 4                         | 4                     | 1                                 |
| 10                 | 26                       | 7                     | 7                               | 7                            | 5                   | 2                        | 1                                        | 1                         | 2                     | 1                                 |
| 11                 | 20                       | 6                     | 2                               | 5                            | 4                   | 1                        | 0                                        | 2                         | 0                     | 0                                 |
| 12                 | 6                        | 7                     | 3                               | 3                            | 9                   | 4                        | 1                                        | 0                         | 1                     | 0                                 |
| 13                 | 10                       | 5                     | 2                               | 4                            | 2                   | 4                        | 1                                        | 1                         | 3                     | 0                                 |
| 14                 | 6                        | 5                     | 2                               | 3                            | 1                   | 1                        | 0                                        | 0                         | 2                     | 0                                 |
| 15                 | 6                        | 1                     | 1                               | 1                            | 3                   | 1                        | 0                                        | 1                         | 0                     | 1                                 |
| 16                 | 6                        | 2                     | 1                               | 1                            | 3                   | 1                        | 0                                        | 2                         | 1                     | 0                                 |
| 17                 | 4                        | 3                     | 2                               | 1                            | 0                   | 1                        | 1                                        | 1                         | 2                     | 0                                 |
| 18                 | 1                        | 1                     | 1                               | 0                            | 2                   | 2                        | 0                                        | 0                         | 1                     | 0                                 |
| 19                 | 3                        | 0                     | 0                               | 1                            | 0                   | 0                        | 0                                        | 1                         | 0                     | 0                                 |
| 20                 | 3                        | 0                     | 1                               | 1                            | 2                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 21                 | 1                        | 0                     | 2                               | 1                            | 1                   | 1                        | 0                                        | 0                         | 2                     | 0                                 |
| 22                 | 2                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 23                 | 1                        | 0                     | 0                               | 0                            | 1                   | 0                        | 1                                        | 0                         | 0                     | 0                                 |
| 24                 | 0                        | 1                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 1                     | 0                                 |
| 25                 | 2                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 26                 | 0                        | 1                     | 0                               | 0                            | 1                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 27                 | 1                        | 0                     | 0                               | 1                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 28                 | 0                        | 0                     | 1                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 29                 | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 2                         | 0                     | 0                                 |
| 30                 | 0                        | 0                     | 1                               | 1                            | 0                   | 0                        | 0                                        | 1                         | 0                     | 0                                 |
| 31                 | 2                        | 0                     | 0                               | 0                            | 1                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 36                 | 1                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 38                 | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 1                                        | 0                         | 0                     | 0                                 |
| 40                 | 0                        | 0                     | 0                               | 1                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 41                 | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 1                     | 0                                 |
| 42                 | 0                        | 0                     | 1                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 45                 | 0                        | 0                     | 0                               | 0                            | 1                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 111                | 1                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |

Table 1b - Effective method line count distribution (normalized)

The table shown here only includes the first 40 rows which typically represents the most significant entries. Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName       | axon-messaging-5.0.3.jar | axon-common-5.0.3.jar | axon-server-connector-5.0.3.jar | axon-eventsourcing-5.0.3.jar | axon-test-5.0.3.jar | axon-modelling-5.0.3.jar | axon-spring-boot-autoconfigure-5.0.3.jar | axon-conversion-5.0.3.jar | axon-update-5.0.3.jar | ax metr micromel 5.0.3 |
|--------------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|------------------------|
| effectiveLineCount |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                        |
| 1                  | 53.560543                | 45.895522             | 41.56250                        | 50.588235                    | 47.855530           | 44.021739                | 59.615385                                | 56.585366                 | 57.142857             | 42.592                 |
| 2                  | 14.519257                | 18.034826             | 22.34375                        | 12.605042                    | 16.478555           | 14.673913                | 23.717949                                | 10.731707                 | 6.493506              | 16.666                 |
| 3                  | 11.748407                | 12.064677             | 12.03125                        | 9.747899                     | 8.126411            | 7.608696                 | 5.128205                                 | 7.804878                  | 3.896104              | 18.518                 |
| 4                  | 5.292325                 | 4.726368              | 6.87500                         | 8.907563                     | 9.029345            | 9.510870                 | 1.923077                                 | 3.902439                  | 6.493506              | 7.407                  |
| 5                  | 4.405653                 | 4.477612              | 5.15625                         | 5.210084                     | 1.354402            | 5.978261                 | 2.564103                                 | 4.878049                  | 5.194805              | 1.851                  |
| 6                  | 2.576891                 | 4.477612              | 2.34375                         | 2.689076                     | 3.386005            | 4.619565                 | 2.564103                                 | 3.414634                  | 5.194805              | 3.703                  |
| 7                  | 2.105846                 | 1.865672              | 1.40625                         | 2.352941                     | 2.031603            | 2.445652                 | 1.602564                                 | 1.951220                  | 1.298701              | 2.777                  |
| 8                  | 1.884178                 | 2.238806              | 2.34375                         | 1.008403                     | 1.805869            | 2.989130                 | 0.641026                                 | 2.926829                  | 1.298701              | 3.703                  |
| 9                  | 1.080632                 | 1.368159              | 1.71875                         | 1.680672                     | 1.805869            | 3.260870                 | 0.320513                                 | 1.951220                  | 2.597403              | 0.925                  |
| 10                 | 0.720421                 | 0.870647              | 1.09375                         | 1.176471                     | 1.128668            | 0.543478                 | 0.320513                                 | 0.487805                  | 1.298701              | 0.925                  |
| 11                 | 0.554170                 | 0.746269              | 0.31250                         | 0.840336                     | 0.902935            | 0.271739                 | 0.000000                                 | 0.975610                  | 0.000000              | 0.000                  |
| 12                 | 0.166251                 | 0.870647              | 0.46875                         | 0.504202                     | 2.031603            | 1.086957                 | 0.320513                                 | 0.000000                  | 0.649351              | 0.000                  |
| 13                 | 0.277085                 | 0.621891              | 0.31250                         | 0.672269                     | 0.451467            | 1.086957                 | 0.320513                                 | 0.487805                  | 1.948052              | 0.000                  |
| 14                 | 0.166251                 | 0.621891              | 0.31250                         | 0.504202                     | 0.225734            | 0.271739                 | 0.000000                                 | 0.000000                  | 1.298701              | 0.000                  |
| 15                 | 0.166251                 | 0.124378              | 0.15625                         | 0.168067                     | 0.677201            | 0.271739                 | 0.000000                                 | 0.487805                  | 0.000000              | 0.925                  |
| 16                 | 0.166251                 | 0.248756              | 0.15625                         | 0.168067                     | 0.677201            | 0.271739                 | 0.000000                                 | 0.975610                  | 0.649351              | 0.000                  |
| 17                 | 0.110834                 | 0.373134              | 0.31250                         | 0.168067                     | 0.000000            | 0.271739                 | 0.320513                                 | 0.487805                  | 1.298701              | 0.000                  |
| 18                 | 0.027709                 | 0.124378              | 0.15625                         | 0.000000                     | 0.451467            | 0.543478                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.000                  |
| 19                 | 0.083126                 | 0.000000              | 0.00000                         | 0.168067                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.487805                  | 0.000000              | 0.000                  |
| 20                 | 0.083126                 | 0.000000              | 0.15625                         | 0.168067                     | 0.451467            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 21                 | 0.027709                 | 0.000000              | 0.31250                         | 0.168067                     | 0.225734            | 0.271739                 | 0.000000                                 | 0.000000                  | 1.298701              | 0.000                  |
| 22                 | 0.055417                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 23                 | 0.027709                 | 0.000000              | 0.00000                         | 0.000000                     | 0.225734            | 0.000000                 | 0.320513                                 | 0.000000                  | 0.000000              | 0.000                  |
| 24                 | 0.000000                 | 0.124378              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.000                  |
| 25                 | 0.055417                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 26                 | 0.000000                 | 0.124378              | 0.00000                         | 0.000000                     | 0.225734            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 27                 | 0.027709                 | 0.000000              | 0.00000                         | 0.168067                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 28                 | 0.000000                 | 0.000000              | 0.15625                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 29                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.975610                  | 0.000000              | 0.000                  |
| 30                 | 0.000000                 | 0.000000              | 0.15625                         | 0.168067                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.487805                  | 0.000000              | 0.000                  |
| 31                 | 0.055417                 | 0.000000              | 0.00000                         | 0.000000                     | 0.225734            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 36                 | 0.027709                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 38                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.320513                                 | 0.000000                  | 0.000000              | 0.000                  |
| 40                 | 0.000000                 | 0.000000              | 0.00000                         | 0.168067                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 41                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.000                  |
| 42                 | 0.000000                 | 0.000000              | 0.15625                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 45                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.225734            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |
| 111                | 0.027709                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                  |

Table 1b Chart 1 - Effective method line count distribution (normalized)

<Figure size 640x480 with 0 Axes>

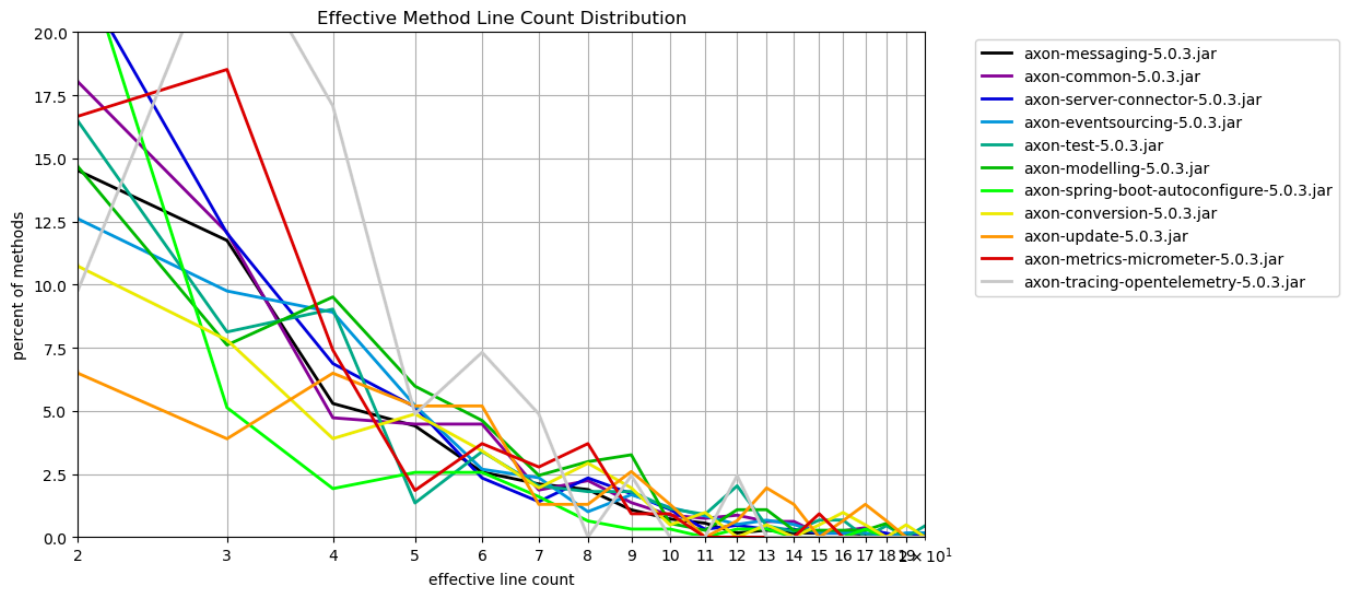


Table 1c - Top 30 packages with highest effective line counts

The following table shows the top 30 packages with the highest effective lines of code. The whole table can be found in the CSV report

`Effective_lines_of_method_code_per_package` .

|    | artifactName                         | fullNamePackage                                   | linesInPackage | methodCount | maxLinesMethod | maxLinesMethod                  |
|----|--------------------------------------|---------------------------------------------------|----------------|-------------|----------------|---------------------------------|
| 0  | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 1409           | 397         | 111            |                                 |
| 1  | axon-messaging-5.0.3                 | org.axonframework.messaging.core                  | 1119           | 531         | 16             |                                 |
| 2  | axon-test-5.0.3                      | org.axonframework.test.fixture                    | 752            | 211         | 45             | ap                              |
| 3  | axon-common-5.0.3                    | org.axonframework.common.configuration            | 710            | 288         | 26             | invo                            |
| 4  | axon-messaging-5.0.3                 | org.axonframework.messaging.core.annotation       | 702            | 235         | 23             |                                 |
| 5  | axon-server-connector-5.0.3          | org.axonframework.axonserver.connector            | 698            | 278         | 42             |                                 |
| 6  | axon-common-5.0.3                    | org.axonframework.common                          | 614            | 183         | 24             | getExactDirectSuperTypesOfParar |
| 7  | axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.eventstore        | 605            | 234         | 19             |                                 |
| 8  | axon-server-connector-5.0.3          | org.axonframework.axonserver.connector.event      | 540            | 162         | 21             | lambd                           |
| 9  | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 436            | 109         | 25             |                                 |
| 10 | axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.eventstore.jpa    | 434            | 136         | 20             |                                 |
| 11 | axon-messaging-5.0.3                 | org.axonframework.messaging.queryhandling         | 381            | 172         | 14             |                                 |
| 12 | axon-messaging-5.0.3                 | org.axonframework.messaging.core.unitofwork       | 375            | 186         | 21             |                                 |
| 13 | axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.springboot.autoconfig | 327            | 145         | 38             |                                 |
| 14 | axon-server-connector-5.0.3          | org.axonframework.axonserver.connector.query      | 325            | 90          | 21             | co                              |
| 15 | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 319            | 121         | 11             |                                 |
| 16 | axon-test-5.0.3                      | org.axonframework.test.matchers                   | 318            | 101         | 21             |                                 |
| 17 | axon-modelling-5.0.3                 | org.axonframework.modelling.entity.annotation     | 307            | 73          | 18             | createOptic                     |
| 18 | axon-conversion-5.0.3                | org.axonframework.conversion.avro                 | 285            | 80          | 30             |                                 |
| 19 | axon-modelling-5.0.3                 | org.axonframework.modelling.entity                | 268            | 76          | 18             |                                 |
| 20 | axon-messaging-5.0.3                 | org.axonframework.messaging.core.configuration    | 256            | 116         | 20             | i                               |
| 21 | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 253            | 94          | 19             |                                 |
| 22 | axon-metrics-micrometer-5.0.3        | org.axonframework.extension.metrics.micrometer    | 252            | 88          | 15             | constructEver                   |
| 23 | axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.springboot            | 248            | 136         | 9              |                                 |
| 24 | axon-common-5.0.3                    | org.axonframework.common.caching                  | 234            | 85          | 13             | lambdacom                       |
| 25 | axon-common-5.0.3                    | org.axonframework.common.infra                    | 233            | 56          | 14             |                                 |
| 26 | axon-messaging-5.0.3                 | org.axonframework.messaging.core.interception     | 217            | 93          | 14             |                                 |
| 27 | axon-messaging-                      | org.axonframework.messaging.core.timeout          | 217            | 56          | 16             |                                 |

|    | artifactName         | fullPackageName                                   | linesInPackage | methodCount | maxLinesMethod | maxL |
|----|----------------------|---------------------------------------------------|----------------|-------------|----------------|------|
|    | 5.0.3                |                                                   |                |             |                |      |
| 28 | axon-messaging-5.0.3 | org.axonframework.messaging.commandhandling       | 216            | 100         | 18             |      |
| 29 | axon-messaging-5.0.3 | org.axonframework.messaging.eventhandling.proc... | 215            | 64          | 19             | lan  |

## Table 1d - Top 30 methods with the highest effective line count

The following table shows the top 30 methods with the highest effective lines of code. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

| index | artifactName                            | fullPackageName                                   | maxLinesMethodType                                 |                           |
|-------|-----------------------------------------|---------------------------------------------------|----------------------------------------------------|---------------------------|
| 0     | 0 axon-messaging-5.0.3                  | org.axonframework.messaging.eventhandling.proc... | Coordinator\$CoordinationTask                      |                           |
| 1     | 2 axon-test-5.0.3                       | org.axonframework.test.fixture                    | Reporter                                           |                           |
| 2     | 5 axon-server-connector-5.0.3           | org.axonframework.axonserver.connector            | AxonServerConnectionManager\$Builder               |                           |
| 3     | 42 axon-update-5.0.3                    | org.axonframework.update.detection                | TestEnvironmentDetector                            |                           |
| 4     | 33 axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.annotation.ref... | AnnotationBasedEventSourcedEntityFactory           | ac                        |
| 5     | 13 axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.springboot.autoconfig | ConverterAutoConfiguration                         |                           |
| 6     | 61 axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.annotation        | AnnotationBasedEventCriteriaResolver\$WrappedEv... |                           |
| 7     | 18 axon-conversion-5.0.3                | org.axonframework.conversion.avro                 | AvroConverter                                      |                           |
| 8     | 67 axon-conversion-5.0.3                | org.axonframework.conversion.jackson              | JacksonConverter                                   |                           |
| 9     | 68 axon-conversion-5.0.3                | org.axonframework.conversion.jackson2             | Jackson2Converter                                  |                           |
| 10    | 3 axon-common-5.0.3                     | org.axonframework.common.configuration            | DefaultAxonApplication\$AxonConfigurationImpl      |                           |
| 11    | 9 axon-messaging-5.0.3                  | org.axonframework.messaging.eventhandling.proc... | JdbcTokenStore                                     |                           |
| 12    | 6 axon-common-5.0.3                     | org.axonframework.common                          | TypeReflectionUtils                                | getExactDirectSuperTypesC |
| 13    | 86 axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.springboot.util       | AbstractQualifiedBeanCondition                     |                           |
| 14    | 4 axon-messaging-5.0.3                  | org.axonframework.messaging.core.annotation       | MethodInvokingMessageHandlingMember                |                           |
| 15    | 40 axon-update-5.0.3                    | org.axonframework.update                          | UpdateCheckerHttpClient                            |                           |
| 16    | 53 axon-modelling-5.0.3                 | org.axonframework.modelling.annotation            | AnnotationBasedEntityEvolvingComponent             |                           |
| 17    | 14 axon-server-connector-5.0.3          | org.axonframework.axonserver.connector.query      | QueryConverter                                     |                           |
| 18    | 16 axon-test-5.0.3                      | org.axonframework.test.matchers                   | DeepEqualsMatcher                                  |                           |
| 19    | 12 axon-messaging-5.0.3                 | org.axonframework.messaging.core.unitofwork       | UnitOfWork\$UnitOfWorkProcessingContext            |                           |
| 20    | 8 axon-server-connector-5.0.3           | org.axonframework.axonserver.connector.event      | AggregateBasedAxonServerEventStorageEngine         | I                         |
| 21    | 46 axon-update-5.0.3                    | org.axonframework.update.api                      | UpdateCheckResponse                                |                           |
| 22    | 35 axon-server-connector-5.0.3          | org.axonframework.axonserver.connector.command    | CommandConverter                                   |                           |
| 23    | 20 axon-messaging-5.0.3                 | org.axonframework.messaging.core.configuration    | MessagingConfigurationDefaults                     |                           |
| 24    | 10 axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.eventstore.jpa    | GapAwareTrackingTokenOperations                    |                           |
| 25    | 32 axon-test-5.0.3                      | org.axonframework.test.server                     | AxonServerContainerUtils                           |                           |
| 26    | 21 axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | TrackerStatus                                      |                           |
| 27    | 29 axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | JpaTokenStore                                      |                           |

| index | artifactName            | fullPackageName                           | maxLinesMethodType         |
|-------|-------------------------|-------------------------------------------|----------------------------|
| 28    | axon-eventourcing-5.0.3 | org.axonframework.eventourcing.eventstore | AnnotationBasedTagResolver |
| 29    | axon-modelling-5.0.3    | org.axonframework.modelling.entity        | ConcreteEntityMetamodel    |

## Cyclomatic Complexity

Table 2a - Cyclomatic method complexity distribution

This table shows the distribution of the cyclomatic complexity of methods per artifact. For each artifact the number of methods with the cyclomatic complexity = 1,2,3,... is shown to get an overview of how cyclomatic complexity is distributed over methods.

Only the 15 artifacts with the highest method count sum and their cyclomatic method complexity distribution (limited by 40) is shown here. The whole table can be found in the CSV report [Cyclomatic\\_Method\\_Complexity\\_Distribution](#).

Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName         | axon-messaging-5.0.3.jar | axon-common-5.0.3.jar | axon-server-connector-5.0.3.jar | axon-eventourcing-5.0.3.jar | axon-test-5.0.3.jar | axon-modelling-5.0.3.jar | axon-spring-boot-autoconfigure-5.0.3.jar | axon-conversion-5.0.3.jar | axon-update-5.0.3.jar | ax metr microme 5.0.3 |
|----------------------|--------------------------|-----------------------|---------------------------------|-----------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------|
| cyclomaticComplexity |                          |                       |                                 |                             |                     |                          |                                          |                           |                       |                       |
| 1                    | 2968                     | 566                   | 536                             | 457                         | 331                 | 291                      | 288                                      | 158                       | 119                   |                       |
| 2                    | 312                      | 95                    | 57                              | 59                          | 41                  | 26                       | 11                                       | 14                        | 16                    |                       |
| 3                    | 160                      | 75                    | 26                              | 37                          | 36                  | 21                       | 7                                        | 9                         | 5                     |                       |
| 4                    | 85                       | 30                    | 7                               | 18                          | 11                  | 18                       | 2                                        | 13                        | 1                     |                       |
| 5                    | 33                       | 10                    | 5                               | 8                           | 8                   | 7                        | 0                                        | 2                         | 3                     |                       |
| 6                    | 34                       | 8                     | 5                               | 6                           | 6                   | 3                        | 1                                        | 2                         | 4                     |                       |
| 7                    | 9                        | 6                     | 1                               | 4                           | 3                   | 1                        | 1                                        | 2                         | 2                     |                       |
| 8                    | 3                        | 8                     | 0                               | 4                           | 4                   | 1                        | 0                                        | 1                         | 0                     |                       |
| 9                    | 1                        | 3                     | 2                               | 0                           | 1                   | 0                        | 1                                        | 1                         | 1                     |                       |
| 10                   | 0                        | 1                     | 0                               | 0                           | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 11                   | 1                        | 1                     | 0                               | 1                           | 1                   | 0                        | 1                                        | 0                         | 1                     |                       |
| 12                   | 0                        | 1                     | 1                               | 0                           | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 13                   | 1                        | 0                     | 0                               | 1                           | 1                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 17                   | 0                        | 0                     | 0                               | 0                           | 0                   | 0                        | 0                                        | 2                         | 1                     |                       |
| 21                   | 0                        | 0                     | 0                               | 0                           | 0                   | 0                        | 0                                        | 1                         | 0                     |                       |
| 25                   | 1                        | 0                     | 0                               | 0                           | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 41                   | 0                        | 0                     | 0                               | 0                           | 0                   | 0                        | 0                                        | 0                         | 1                     |                       |
| 58                   | 1                        | 0                     | 0                               | 0                           | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |

Table 2b - Cyclomatic method complexity distribution (normalized)

The table shown here only includes the first 40 rows which typically represents the most significant entries. Have a look below to find out which packages and methods have the



highest effective lines of code.

| artifactName         | axon-messaging-5.0.3.jar | axon-common-5.0.3.jar | axon-server-connector-5.0.3.jar | axon-eventsourcing-5.0.3.jar | axon-test-5.0.3.jar | axon-modelling-5.0.3.jar | axon-spring-boot-autoconfigure-5.0.3.jar | axon-conversion-5.0.3.jar | axon-update-5.0.3.jar | axon-metrics-micrometer-5.0.3.jar | axon-tracing-opentelemetry-5.0.3.jar |
|----------------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------------------|--------------------------------------|
| cyclomaticComplexity |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                                   |                                      |
| 1                    | 82.238847                | 70.398010             | 83.75000                        | 76.806723                    | 74.717833           | 79.076087                | 92.307692                                | 77.073171                 | 77.272727             | 88.8                              |                                      |
| 2                    | 8.645054                 | 11.815920             | 8.90625                         | 9.915966                     | 9.255079            | 7.065217                 | 3.525641                                 | 6.829268                  | 10.389610             | 9.7                               |                                      |
| 3                    | 4.433361                 | 9.328358              | 4.06250                         | 6.218487                     | 8.126411            | 5.706522                 | 2.243590                                 | 4.390244                  | 3.246753              | 1.8                               |                                      |
| 4                    | 2.355223                 | 3.731343              | 1.09375                         | 3.025210                     | 2.483070            | 4.891304                 | 0.641026                                 | 6.341463                  | 0.649351              | 0.0                               |                                      |
| 5                    | 0.914381                 | 1.243781              | 0.78125                         | 1.344538                     | 1.805869            | 1.902174                 | 0.000000                                 | 0.975610                  | 1.948052              | 0.0                               |                                      |
| 6                    | 0.942089                 | 0.995025              | 0.78125                         | 1.008403                     | 1.354402            | 0.815217                 | 0.320513                                 | 0.975610                  | 2.597403              | 0.0                               |                                      |
| 7                    | 0.249377                 | 0.746269              | 0.15625                         | 0.672269                     | 0.677201            | 0.271739                 | 0.320513                                 | 0.975610                  | 1.298701              | 0.0                               |                                      |
| 8                    | 0.083126                 | 0.995025              | 0.00000                         | 0.672269                     | 0.902935            | 0.271739                 | 0.000000                                 | 0.487805                  | 0.000000              | 0.0                               |                                      |
| 9                    | 0.027709                 | 0.373134              | 0.31250                         | 0.000000                     | 0.225734            | 0.000000                 | 0.320513                                 | 0.487805                  | 0.649351              | 0.0                               |                                      |
| 10                   | 0.000000                 | 0.124378              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |                                      |
| 11                   | 0.027709                 | 0.124378              | 0.00000                         | 0.168067                     | 0.225734            | 0.000000                 | 0.320513                                 | 0.000000                  | 0.649351              | 0.0                               |                                      |
| 12                   | 0.000000                 | 0.124378              | 0.15625                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |                                      |
| 13                   | 0.027709                 | 0.000000              | 0.00000                         | 0.168067                     | 0.225734            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |                                      |
| 17                   | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.975610                  | 0.649351              | 0.0                               |                                      |
| 21                   | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.487805                  | 0.000000              | 0.0                               |                                      |
| 25                   | 0.027709                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |                                      |
| 41                   | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.0                               |                                      |
| 58                   | 0.027709                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |                                      |

Table 2b Chart 1 - Cyclomatic method complexity distribution (normalized)

<Figure size 640x480 with 0 Axes>

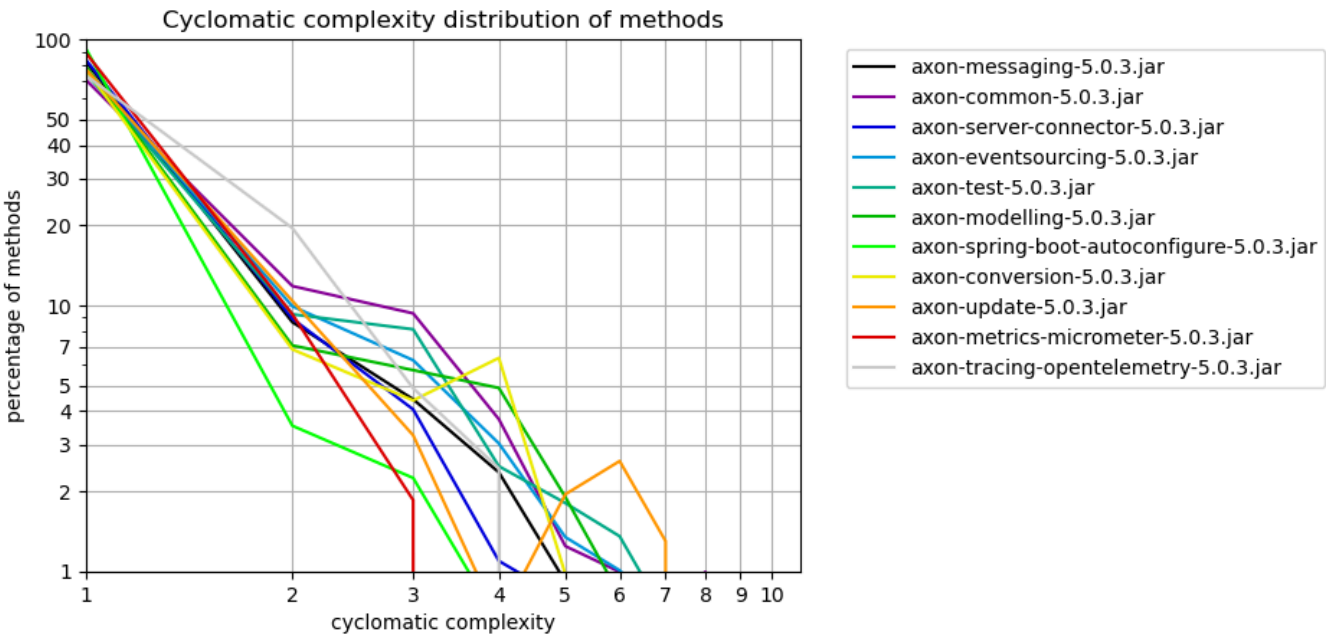


Table 2c - Top 30 packages with highest cyclomatic complexity

The following table shows the top 30 packages with the highest cyclomatic complexity. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

|    | artifactName                         | fullPackageName                                   | complexityInPackage | methodCount | maxComplexity | maxComplexityMethod            |
|----|--------------------------------------|---------------------------------------------------|---------------------|-------------|---------------|--------------------------------|
| 1  | axon-messaging-5.0.3                 | org.axonframework.messaging.core                  | 731                 | 531         | 9             | process                        |
| 0  | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 548                 | 397         | 25            | run                            |
| 6  | axon-common-5.0.3                    | org.axonframework.common                          | 388                 | 183         | 9             | fieldNameFromMember            |
| 3  | axon-common-5.0.3                    | org.axonframework.common.configuration            | 388                 | 288         | 8             | hasComponent                   |
| 7  | axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.eventstore        | 377                 | 234         | 8             | from                           |
| 4  | axon-messaging-5.0.3                 | org.axonframework.messaging.core.annotation       | 371                 | 235         | 13            | resolveToStream                |
| 5  | axon-server-connector-5.0.3          | org.axonframework.axonserver.connector            | 357                 | 278         | 12            | build                          |
| 2  | axon-test-5.0.3                      | org.axonframework.test.fixture                    | 325                 | 211         | 11            | appendEventOverview            |
| 15 | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 248                 | 121         | 7             | covers                         |
| 12 | axon-messaging-5.0.3                 | org.axonframework.messaging.core.unitofwork       | 242                 | 186         | 6             | commit                         |
| 8  | axon-server-connector-5.0.3          | org.axonframework.axonserver.connector.event      | 218                 | 162         | 4             | appendEvents                   |
| 11 | axon-messaging-5.0.3                 | org.axonframework.messaging.queryhandling         | 216                 | 172         | 4             | subscribe                      |
| 10 | axon-eventsourcing-5.0.3             | org.axonframework.eventsourcing.eventstore.jpa    | 207                 | 136         | 8             | withGapsCleaned                |
| 9  | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 194                 | 109         | 8             | updateToken                    |
| 13 | axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.springboot.autoconfig | 191                 | 145         | 11            | buildConverter                 |
| 16 | axon-test-5.0.3                      | org.axonframework.test.matchers                   | 173                 | 101         | 8             | matchingFields                 |
| 18 | axon-conversion-5.0.3                | org.axonframework.conversion.avro                 | 149                 | 80          | 21            | convert                        |
| 21 | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.proc... | 147                 | 94          | 8             | equals                         |
| 24 | axon-common-5.0.3                    | org.axonframework.common.caching                  | 147                 | 85          | 8             | onEvent                        |
| 34 | axon-messaging-5.0.3                 | org.axonframework.messaging.eventhandling.anno... | 144                 | 56          | 58            | parseParameter                 |
| 23 | axon-spring-boot-autoconfigure-5.0.3 | org.axonframework.extension.springboot            | 139                 | 136         | 2             | batchSize                      |
| 20 | axon-messaging-5.0.3                 | org.axonframework.messaging.core.configuration    | 129                 | 116         | 4             | lambda<br>registerDecorators15 |
| 14 | axon-server-connector-5.0.3          | org.axonframework.axonserver.connector.query      | 127                 | 90          | 6             | sendResponses                  |
| 17 | axon-modelling-5.0.3                 | org.axonframework.modelling.entity.annotation     | 124                 | 73          | 6             | getExpectedRepresentation      |
| 19 | axon-modelling-5.0.3                 | org.axonframework.modelling.entity                | 119                 | 76          | 7             | handleInstance                 |
| 43 | axon-messaging-5.0.3                 | org.axonframework.messaging.eventstreaming        | 119                 | 92          | 4             | havingTags                     |
| 28 | axon-messaging-5.0.3                 | org.axonframework.messaging.commandhandling       | 117                 | 100         | 4             | subscribe                      |

|    | artifactName         | fullPackageName                                   | complexityInPackage | methodCount | maxComplexity | maxComplexityMethod |
|----|----------------------|---------------------------------------------------|---------------------|-------------|---------------|---------------------|
| 26 | axon-messaging-5.0.3 | org.axonframework.messaging.core.interception     | 107                 | 93          | 4             | convert             |
| 29 | axon-messaging-5.0.3 | org.axonframework.messaging.eventhandling.proc... | 106                 | 64          | 6             | equals              |
| 30 | axon-messaging-5.0.3 | org.axonframework.messaging.eventhandling         | 106                 | 86          | 4             | handle              |

## Table 2d - Top 30 methods with highest cyclomatic complexity

The following table shows the top 30 packages containing the methods with the highest cyclomatic complexity. The whole table can be found in the CSV report

`Effective_lines_of_method_code_per_package` .

| index | artifactName | fullPackageName                                                                           | maxComplexityType                                  | maxComplexityMe          |
|-------|--------------|-------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------|
| 0     | 34           | axon-messaging-5.0.3<br>org.axonframework.messaging.eventhandling.anno...                 | MethodSequencingPolicyEventHandlerDefinition\$S... | parseParar               |
| 1     | 42           | axon-update-5.0.3<br>org.axonframework.update.detection                                   | TestEnvironmentDetector                            | isTest                   |
| 2     | 0            | axon-messaging-5.0.3<br>org.axonframework.messaging.eventhandling.proc...                 | Coordinator\$CoordinationTask                      |                          |
| 3     | 18           | axon-conversion-5.0.3<br>org.axonframework.conversion.avro                                | AvroConverter                                      | co                       |
| 4     | 67           | axon-conversion-5.0.3<br>org.axonframework.conversion.jackson                             | JacksonConverter                                   | co                       |
| 5     | 46           | axon-update-5.0.3<br>org.axonframework.update.api                                         | UpdateCheckResponse                                | fromRe                   |
| 6     | 68           | axon-conversion-5.0.3<br>org.axonframework.conversion.jackson2                            | Jackson2Converter                                  | co                       |
| 7     | 4            | axon-messaging-5.0.3<br>org.axonframework.messaging.core.annotation                       | MessageStreamResolverUtils                         | resolveToSt              |
| 8     | 33           | axon-eventourcing-5.0.3<br>org.axonframework.eventsourcing.annotation.ref...              | AnnotationBasedEventSourcedEntityFactory           | addEntityCreatorExecu    |
| 9     | 32           | axon-test-5.0.3<br>org.axonframework.test.server                                          | AxonServerContainer                                | e                        |
| 10    | 48           | axon-common-5.0.3<br>org.axonframework.common.jdbc                                        | ConnectionWrapperFactory                           | lambdaW                  |
| 11    | 5            | axon-server-connector-5.0.3<br>org.axonframework.axonserver.connector                     | AxonServerConnectionManager\$Builder               |                          |
| 12    | 13           | axon-spring-boot-autoconfigure-5.0.3<br>org.axonframework.extension.springboot.autoconfig | ConverterAutoConfiguration                         | buildConv                |
| 13    | 2            | axon-test-5.0.3<br>org.axonframework.test.fixture                                         | Reporter                                           | appendEventOve           |
| 14    | 61           | axon-eventourcing-5.0.3<br>org.axonframework.eventsourcing.annotation                     | AnnotationBasedEventCriteriaResolver\$WrappedEv... | <                        |
| 15    | 52           | axon-common-5.0.3<br>org.axonframework.common.lock                                        | PessimisticLockFactory\$DisposableLock             |                          |
| 16    | 6            | axon-common-5.0.3<br>org.axonframework.common                                             | ReflectionUtils                                    | fieldNameFromMe          |
| 17    | 1            | axon-messaging-5.0.3<br>org.axonframework.messaging.core                                  | MessageStreamUtils\$Reducer                        | pr                       |
| 18    | 24           | axon-common-5.0.3<br>org.axonframework.common.caching                                     | EhCacheAdapter\$CacheEventListenerAdapter          | onl                      |
| 19    | 70           | axon-common-5.0.3<br>org.axonframework.common.annotation                                  | AnnotationUtils                                    | collectAttri             |
| 20    | 37           | axon-modelling-5.0.3<br>org.axonframework.modelling.entity.child                          | FieldChildEntityFieldDefinition                    | evolveParentBasedOnChild |
| 21    | 60           | axon-test-5.0.3<br>org.axonframework.test.extension                                       | ProvidedAxonTestFixtureUtils                       | validateValueL           |
| 22    | 9            | axon-messaging-5.0.3<br>org.axonframework.messaging.eventhandling.proc...                 | JdbcTokenStore                                     | update                   |
| 23    | 21           | axon-messaging-5.0.3<br>org.axonframework.messaging.eventhandling.proc...                 | TrackerStatus                                      | e                        |
| 24    | 3            | axon-common-5.0.3<br>org.axonframework.common.configuration                               | DefaultComponentRegistry                           | hasCompr                 |
| 25    | 7            | axon-eventourcing-5.0.3<br>org.axonframework.eventsourcing.eventstore                     | AggregateBasedConsistencyMarker                    |                          |
| 26    | 10           | axon-eventourcing-5.0.3<br>org.axonframework.eventsourcing.eventstore.jpa                 | GapAwareTrackingTokenOperations                    | withGapsCle              |
| 27    | 16           | axon-test-5.0.3<br>org.axonframework.test.matchers                                        | DeepEqualsMatcher                                  | matchingI                |
| 28    | 77           | axon-common-5.0.3<br>org.axonframework.common.util                                        | PriorityRunnable                                   | e                        |

| index | artifactName                | fullPackageName                | maxComplexityType                                | maxComplexityMe |
|-------|-----------------------------|--------------------------------|--------------------------------------------------|-----------------|
| 29    | 25<br>axon-<br>common-5.0.3 | org.axonframework.common.infra | FilesystemStyleComponentDescriptor\$TreeRenderer | renderPro       |